ORIE 5370

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1 Extending Linear IPO to Nonlinear IPO

1.1 Motivation

The original integrated prediction and optimization (IPO) framework considers a linear prediction model for expected asset returns and optimizes the prediction parameters directly with respect to downstream mean-variance portfolio performance. This is attractive because it aligns the training objective with the final investment decision rather than with a standalone prediction loss.

However, linear predictors may be too restrictive in practice. Asset returns often depend on features in a nonlinear manner, through interactions, higher-order effects, or more flexible latent representations. Therefore, a natural extension is to replace the linear return predictor with a nonlinear function while preserving the decision-focused IPO objective. This leads to a nonlinear IPO formulation.

There are two conceptually distinct routes to nonlinear IPO:

1. **Nonlinear in inputs but linear in parameters:** use a nonlinear feature map $\phi(x)$ and define the predictor as a linear model in the transformed features.

2. **Fully nonlinear in parameters:** use a neural network, kernel model, or another differentiable nonlinear predictor $f(x;\theta)$.

The first route preserves much of the analytical structure of the original IPO formulation, while the second generally requires gradient-based optimization through the downstream portfolio problem.

1.2 General Nonlinear IPO Formulation

Let $x \in R^{d_x}$ denote the input features and let $y \in R^{d_y}$ denote the realized asset returns. We consider a general predictor

$$\hat{y} = f(x;\theta),$$

where θ denotes the model parameters.

For each observation $i = 1, \ldots, m$, the downstream portfolio decision is obtained by solving a mean-variance optimization problem using the predicted returns $\hat{y}_i = f(x_i; \theta)$. The nonlinear IPO problem is

$$\min_{\theta} \; \frac{1}{m} \sum_{i=1}^{m} c(z_i^*(f(x_i; \theta)), \, y_i) \tag{1}$$

subject to

$$z_i^*(\hat{y}_i) = \arg\min_{z \in S} \left(-z^\top \hat{y}_i + \frac{\delta}{2} z^\top \hat{V}_i z \right), \tag{2}$$

where $\delta > 0$ is the risk-aversion parameter, $\hat{V}_i$ is the estimated covariance matrix, and S denotes the feasible set of portfolio weights.

The realized downstream cost is

$$c(z, y) = -z^\top y + \frac{\delta}{2} z^\top V z, \tag{3}$$

where V is the true covariance matrix (or an out-of-sample proxy).

1.3 Route I: Nonlinear Features with Parameter-Linear Prediction

A natural extension is to introduce a nonlinear feature map

$$\phi(x) \in R^p,$$

and define the prediction model as

$$\hat{y} = W\phi(x) + b. \tag{4}$$

Examples include polynomial expansions, interaction terms, spline bases, radial basis features, or other engineered nonlinear transformations.

To simplify notation, augment the feature vector with a constant:

$$\tilde{\phi}(x) = \phi(x)1, \qquad \hat{y} = \Theta^\top \tilde{\phi}(x),$$

where Θ stacks W and b.

Although the predictor is nonlinear in the original inputs x, it remains linear in the parameters Θ. Therefore, the analytical structure of the original linear IPO framework can be preserved after replacing x with $\phi(x)$.

1.4 Unconstrained Mean-Variance Problem

We first consider the unconstrained case $S = R^{d_y}$. The inner problem becomes

$$z_i^*(\hat{y}_i) = \arg\min_{z} \left(-z^\top \hat{y}_i + \frac{\delta}{2} z^\top \hat{V}_i z \right).$$

The first-order condition is

$$-\hat{y}_i + \delta \hat{V}_i z = 0,$$

which implies

$$z_i^*(\hat{y}_i) = \frac{1}{\delta} \hat{V}_i^{-1} \hat{y}_i. \tag{5}$$

Substituting the parameter-linear nonlinear predictor $\hat{y}_i = \Theta^\top \tilde{\phi}_i$ into eq:unconstrained$_{o}ptimal_{z}, we obtain z_i^*($
$\frac{1}{\delta} \hat{V}_i^{-1} \Theta^\top \tilde{\phi}_i$.

The realized cost for sample i is

$$\ell_i(\Theta) = -z_i^{*\top} y_i + \frac{\delta}{2} z_i^{*\top} V_i z_i^*.$$

Substituting $z_i^*(\Theta)$ gives $\ell_i(\Theta) = -\frac{1}{\delta} \tilde{\phi}_i^\top \Theta \hat{V}_i^{-1} y_i + \frac{1}{2\delta} \tilde{\phi}_i^\top \Theta \hat{V}_i^{-1} V_i \hat{V}_i^{-1} \Theta^\top \tilde{\phi}_i$.

Therefore, the full nonlinear-feature IPO objective is

$$L(\Theta) = \frac{1}{m} \sum_{i=1}^{m} \ell_i(\Theta), \tag{6}$$

which is a quadratic function of Θ. Hence, this extension remains analytically tractable and can be solved as a convex quadratic optimization problem, or equivalently as a generalized normal equation system.

This shows that *nonlinearity in the input space does not destroy the analytical IPO structure as long as the predictor remains linear in its parameters.*

1.5 Equality-Constrained Mean-Variance Problem

Next, suppose the feasible set is defined by linear equality constraints,

$$S = \{z : Az = b\}.$$

Using a null-space parameterization, any feasible portfolio can be written as

$$z = Fq + z_0,$$

where the columns of F span the null space of A, i.e. $AF = 0$, and z_0 is any feasible point satisfying $Az_0 = b$.

Substituting into the inner problem yields an optimization problem in q. The optimal decision can be expressed as

$$z_i^*(\hat{y}_i) = \frac{1}{\delta} F (F^\top \hat{V}_i F)^{-1} F^\top \hat{y}_i + \left(I - F (F^\top \hat{V}_i F)^{-1} F^\top \hat{V}_i \right) z_0. \tag{7}$$

Define

$$B_i = \frac{1}{\delta} F (F^\top \hat{V}_i F)^{-1} F^\top, \qquad a_i = \left(I - F (F^\top \hat{V}_i F)^{-1} F^\top \hat{V}_i \right) z_0.$$

Then

$$z_i^*(\hat{y}_i) = B_i \hat{y}_i + a_i. \tag{8}$$

Substituting $\hat{y}_i = \Theta^\top \tilde{\phi}_i$ into eq:affine$_decision_rule, the decision becomes affine in \Theta$. The sample loss is

$$\ell_i(\Theta) = -(B_i \Theta^\top \tilde{\phi}_i + a_i)^\top y_i + \frac{\delta}{2}(B_i \Theta^\top \tilde{\phi}_i + a_i)^\top V_i (B_i \Theta^\top \tilde{\phi}_i + a_i).$$

Again, this is quadratic in Θ, so the nonlinear-feature IPO formulation remains analytically solvable in the equality-constrained case.